

MM RIDE - COMPLETE MANUAL TESTING PROTOCOL

VERIFIED READY

End-to-End Operational Lifecycle & Anti-Fraud Verification Checklist

Fleet Setup:

1 Vehicle (Owner-cum-Driver)

Admin Portal:

Admin Web (Vite + React)

Mobile App:

Driver App (Expo / React Native)

Target OS:

Android / iOS / Mobile Browser

📌 Testing Methodology (Dual Role Execution)

Since you have **1 Vehicle** and are acting as both **Owner & Driver**: You will configure the vehicle and approval workflow in the **Admin Web (Owner Portal)**, and perform physical field operations (inspections, meter reading, shifts, GPS telemetry, trips, and daily submission) via the **Driver App (Mobile)**.

1 PHASE 1: OWNER PRE-SETUP (Admin Web Portal)

STATUS	SCREEN / PAGE	ACTION & INPUTS	EXPECTED VERIFICATION RESULT
☐	Hubs Management Owner Web	Click Add Hub : - Name: Central Depot - Coordinates: Lat 13.0827, Lng 80.2707 (or your current location) - Geofence Radius: 500 meters	Hub is saved in Firestore and appears in depot list. Used for geofenced shift start/end checks.
☐	Bikes Fleet Owner Web	Click Add Bike : - Reg Number: TN 09 AB 1234 - Model: Electric / Petrol - Initial Odometer: 1000 km - Battery/Fuel: 100% - Status: AVAILABLE	Bike is registered with available status. Ready for driver assignment.
☐	System Settings Owner Web	Review parameters: - Max Daily Duty: 12 hours - Speed Alert Limit: 60 km/h - Idle Alert Interval: 30 mins	Global configuration verified in Firestore settings/system doc.

2 PHASE 2: DRIVER REGISTRATION & KYC ONBOARDING (Driver App)

STATUS	SCREEN / PAGE	ACTION & INPUTS	EXPECTED VERIFICATION RESULT
☐	Login / Auth Driver App	Open Driver App on phone. Enter phone number or test credentials and authenticate.	Firestore Auth signs in; new profile routes to RegistrationScreen.
☐	Profile Details Driver App	- Full Name: Driver's name - Driving Licence (DL): TN0920230001234 - Address: Auto-fetched GPS address - Emergency Phone: 10-digit number - UPI ID: driver@upi	Form validates mandatory fields. Auto-capitalizes DL. Saves profile with status DOCUMENTS_SUBMITTED.
☐	Document Upload Driver App	Capture/Upload KYC docs: - DL Front & Back photo - Aadhaar Front & Back photo - Live Driver Profile Selfie	Images uploaded to storage. Status transitions to DOCUMENT_VERIFICATION_PENDING.
☐	Status Gate Check Driver App	View VerificationStatusScreen.	In-review banner displayed. Access to Home Screen is locked until Owner approves.

3 PHASE 3: OWNER VERIFICATION & VEHICLE ALLOCATION (Admin Web → Driver App)

STATUS	SCREEN / PAGE	ACTION & INPUTS	EXPECTED VERIFICATION RESULT
☐	Verification Page Owner Web	Open pending driver card, inspect uploaded DL & Aadhaar images, click Approve Driver .	Driver account status updates to APPROVED in Firestore.
☐	Bikes Fleet Page Owner Web	Select the 1 vehicle (TN 09 AB 1234), click Assign Driver , select your driver profile.	Bike status changes to ASSIGNED; Driver record updated with assignedBikeId.
☐	Instant Mobile Sync Driver App	Check phone screen. Real-time Firestore listener auto-detects assignment without manual restart.	Status screen immediately displays: "Bike Assigned - Proceed to Handover" .

4 PHASE 4: 360° VEHICLE INSPECTION & CUSTODY ACCEPTANCE (Driver App)

STATUS	SCREEN / PAGE	ACTION & INPUTS	EXPECTED VERIFICATION RESULT
☐	Meter & Levels Driver App	Verify displayed starting odometer (1000 km) and fuel charge (100%).	Values match the Admin bike ledger accurately.
☐	360° Camera Photos Driver App	Capture 5 mandatory live photos: 1. Front 2. Rear 3. Left 4. Right 5. Meter Console	Camera opens natively (gallery upload blocked for anti-fraud). Thumbnails render.
☐	Hardware & Doc Checks Driver App	Toggle all physical items: ✓ Chassis & Engine No. match RC ✓ Physical RC & Insurance received ✓ Keys received: 1 Helmets: 1 ✓ GPS tracker verified active	All toggle switches become active green.
☐	Custody Sign-Off Driver App	Accept legal responsibility declaration and click Accept Custody & Complete Handover .	Handover record saved. Driver account status set to ACTIVE_DRIVER. HomeScreen unlocked!

5 PHASE 5: SHIFT START & PRE-DUTY CHECKS (Driver App)

STATUS	SCREEN / PAGE	ACTION & INPUTS	EXPECTED VERIFICATION RESULT
☐	Home Screen Status Driver App	Verify status pill:  OFFLINE (Ready to Work). Click Start Work / Shift  .	Opens StartDutyScreen.
☐	Depot Proximity Check Driver App	Select starting Hub from dropdown.	App verifies GPS distance. Green badge confirms driver is within depot geofence.
☐	Odometer Tamper Test Driver App	• Test entering 950 (less than 1000). • Then enter valid 1000.	Alert blocks start: <i>"Meter cannot be less than previous odometer"</i> . Prevents rollback fraud!
☐	Safety Gear & QR Scan Driver App	• Confirm helmet toggle ON. • Tap Scan Bike QR button.	Simulates optical QR match with vehicle plate (TN09AB1234).
☐	Duty Session Kickoff Driver App	Click Start Active Shift  .	Shift session created in Firestore. Background GPS tracking begins. Console opens directly.

6 PHASE 6: LIVE ACTIVE DUTY & ANTI-FRAUD SUPERVISION (Driver App & Admin Web)

STATUS	FEATURE / TEST AREA	EXECUTION STEPS	EXPECTED VERIFICATION RESULT
☐	Live GPS Heartbeat Owner Web	Open Live Monitoring in Admin Web. Check driver pin on map.	Live map pin updates every few seconds with current speed, shift timer, and battery/fuel.
☐	Gig Platform Simulation Driver App	In Active Console, tap Simulate Platform Ride : Select: <i>Request</i> → <i>Accept</i> → <i>Complete</i> .	Logs to platformRideEvents. Driver profile updates with platform ride status in real-time.
☐	Offline Cash Ride Alarm Driver App	Tap Simulate Rider Cancelled while vehicle continues moving.	Admin Web flag triggers: OFFLINE_CASH alert (detects trip cancelled to pocket customer cash).
☐	Remote Live Selfie Challenge Owner Web	In Admin Web Live Monitoring, click Trigger Live Selfie Challenge on the driver.	Driver phone pops a 90-second countdown selfie modal . Driver submits face photo; verified in Admin.
☐	Emergency SOS Incident Driver App	Tap red SOS button in Driver App header.	High-priority emergency incident logged in Admin Web Incidents with live GPS coordinates.
☐	Sudden Deceleration / Accident Driver App	Trigger deceleration simulation (speed drops from >30 to 0 abruptly).	60s Welfare Safety Modal appears: <i>"Are you safe? 1. Safe, 2. Breakdown, 3. Accident"</i> .
☐	Fuel / Petrol Entry Driver App	Tap Log Petrol / Fuel Fill . Enter: ₹200, 1.95L, meter 1020 km, dispenser photo.	Fuel expense saved. Visible in Admin Web EarningsSettlement → Fuel Expenses tab.
☐	Remote Engine Immobilization Owner Web	In Admin Web, toggle Immobilize Engine on the active driver.	Driver app displays full-screen red warning: <i>"ENGINE REMOTELY IMMOBILIZED BY DEPOT"</i> .
☐	Shift Break Flow Driver App	Tap Take a Break → BreakScreen opens. Tap Resume Duty .	Duty status toggles ON_BREAK → ACTIVE smoothly. Shift timer pauses correctly.

7 PHASE 7: SHIFT CONCLUSION & DEPOT RETURN AUDIT (Driver App)

STATUS	SCREEN / PAGE	ACTION & INPUTS	EXPECTED VERIFICATION RESULT
☐	End Shift Trigger Driver App	In Active Console, tap End Shift .	Opens EndDutyScreen. Displays start odometer (1000 km) and hub destination.
☐	Return Odometer Audit Driver App	Enter return reading: 1045 km.	Auto-calculates distance covered: 45 km . Validates reading cannot be < start.
☐	Depot Geofence Validation Driver App	App verifies GPS distance to depot (must be within 500m or emergency override used).	Proximity badge turns green. Off-depot return requires written justification.
☐	Anti-Part Swap Checklist Driver App	Confirm mandatory return checks: ☒ Commercial bike keys handed in ☒ Helmets returned ☒ No battery, tyres, or parts swapped	All toggle switches activated. Submit button becomes enabled.
☐	Duty Completion Driver App	Click Confirm Depot Return & End Shift .	Session marked COMPLETED. Total duty duration calculated. Vehicle returned. Routes to Home.

8 PHASE 8: DAILY EARNINGS SUBMISSION & 50/50 SETTLEMENT AUDIT (Driver App → Admin Web)

STATUS	SCREEN / PAGE	ACTION & INPUTS	EXPECTED VERIFICATION RESULT
☐	Submit Daily Earnings Driver App	Tap Submit Daily Earnings : - Gross Income: ₹1500 - Platform Commission: ₹300 - Completed Rides: 8 Cash Collected: ₹500 • Attach Ola/Uber app summary screenshot	Submission saved with PENDING_REVIEW. Alert confirms submission to Operations.
☐	Admin Submission Audit Owner Web	Open EarningsSettlement → <i>Submissions</i> tab. Inspect uploaded screenshot proof.	Submission displays with image thumbnail, driver ID, and revenue figures. Click Convert to Settlement .
☐	50/50 Split Calculation Owner Web	Verify automated mathematical breakdown: Gross Income: ₹1,500 - Platform Fees: ₹300 = Net Income: ₹1,000 - Driver 50% Share: ₹500 - Owner 50% Share: ₹500 - Less: 10% Maintenance Reserve: -₹50 - Less: Driver Cash Collected: - ₹400 (already with driver) NET PAYABLE TO DRIVER TODAY = ₹50	Formula calculation matches penny-for-penny. No manual spreadsheet errors!
☐	Payout Approval & Record Owner Web	Select Method: UPI, enter transaction ref UPI987654321, click Mark as Paid .	Settlement marked PAID. Financial ledger updated in Firestore.
☐	Driver Earnings Verification Driver App	In Driver App, open Earnings Screen .	Shows today's earnings, 50% share, deductions, and green PAID  status tag.